

Petrographic Analysis of Apollo Lunar Sample 12004-11-r1

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Paired PPL/XPL rotation series for Apollo lunar sample 12004-11-r1 was obtained initially from the Virtual Microscope [1]. These petrographic images were subsequently aligned to ensure a consistent field of view by using Python. The rock sample 12004 was collected on (or near) the rim of Middle Crescent Crater during the Apollo 12 mission.

1 Phenocrysts

1.1 Dominant phase

The bulk of the section is composed of large anhedral to subhedral, equant grains that are bright and non-pleochroic in PPL with moderate-to-high positive relief and no cleavage, instead showing the characteristic pervasive irregular/conchoidal fracturing. In XPL, these grains display high birefringence, with vivid upper-second-to-third-order interference colors (magenta, purple, blue, yellow, green). Through the rotation series, individual grains pass cleanly into and out of extinction at successive stage positions, consistent with the sharp extinction expected of olivine. The combination of bright high relief, lack of cleavage, conchoidal fracture, and high-order interference colors is diagnostic of olivine.

1.2 Blocky brownish grains

Blocky grains with cleavage and second-order interference colors, locally with weak brownish tints, present in roughly equal proportion to olivine among the phenocrysts, with a comparable size range, are tentatively pyroxene.

2 Variolitic groundmass

The groundmass (60% of the rock) is built of plagioclase and pyroxene intergrown as sheaf-like, fan-shaped, sub-radiating bundles (varioles) that nucleate on common centers and on the margins of the larger crystals and splay outward from them. In PPL, the texture reads as feathery sprays of fine, dark, acicular-to-bladed crystals threading between the phenocrysts. In XPL, the bundles carry low-to-moderate (first-order) interference colors and diagnostically show sweeping/undulatory extinction: as the stage rotates, each fan extinguishes progressively from one side

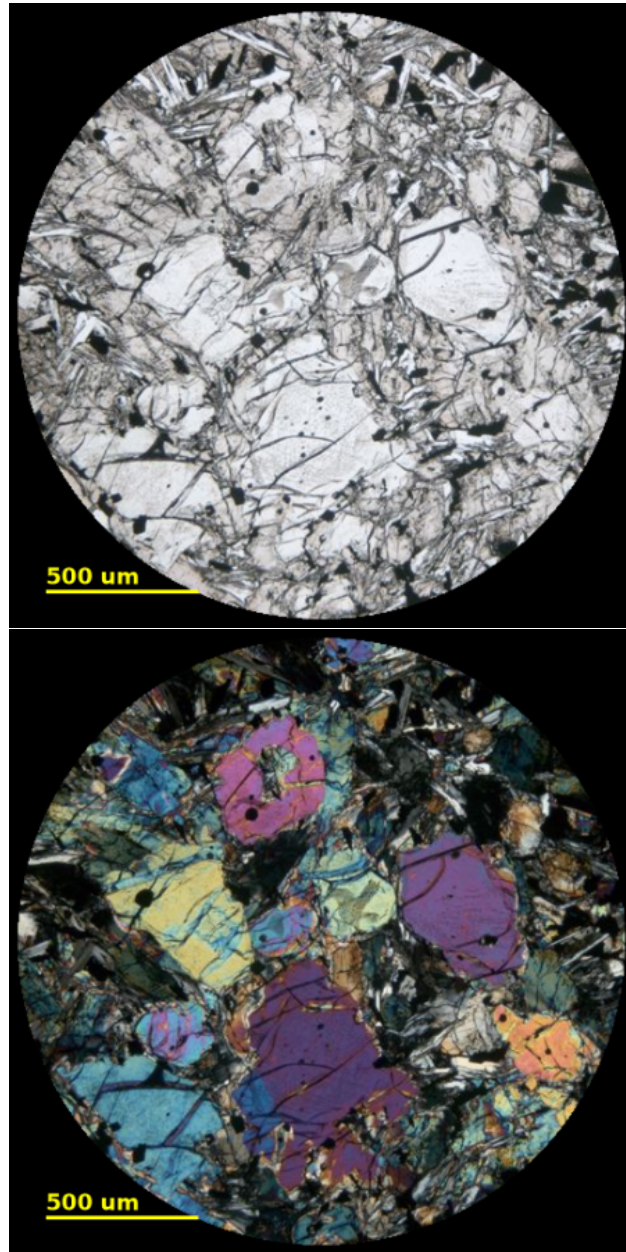


Figure 1: Sample 12004-11-r1 in PPL (top) and XPL (bottom). Field of view 2.0 mm; scale bar 500 μm . Equant olivine and blocky clinopyroxene phenocrysts (high-order interference colors in XPL) sit in a finer variolitic groundmass of plagioclase & pyroxene.

to the other rather than all at once, because its constituent fibers fan across a range of orientations. The plagioclase laths are thin and locally hollow/skeletal, consistent with rapid growth. The texture is appropriately described as variolitic to sub-variolitic [2,3].

3 Opaque and accessory phases

Skeletal to lath-shaped opaque grains are intergrown within the variolitic bundles and along pyroxene margins. Most probably, equant, blocky chromite occurs as discrete grains and as inclusions, locally with brown ulvöspinel rims characteristic of this basalt type; and minor blebs of troilite and native Fe-metal occur in the groundmass and in association with the spinel grains.

4 Interpretation

The order-of-magnitude step from sub-millimeter olivine & clinopyroxene phenocrysts to 10–20 μm groundmass fibers records a two-stage cooling history, typical of a lunar mare lava: slow early growth of olivine and pyroxene, then rapid undercooling that drove variolitic crystallization of the residual plagioclase-bearing melt.

References

- [1] "12004 Olivine basalt," *The Virtual Microscope*, The Open University. [Online]. Available: <https://www.virtualmicroscope.org/content/12004-11-olivine-basalt>
- [2] R. Brett, P. Butler, C. Meyer, A. M. Reid, H. Takeda, and R. Williams, "Apollo 12 igneous rocks 12004, 12008, 12009, and 12022: A mineralogical and petrological study," in *Proceedings of the Second Lunar Science Conference*, Cambridge, MA, USA: MIT Press, 1971, pp. 301–317.
- [3] C. Meyer, *Lunar Sample Compendium*. Houston, TX, USA: NASA Johnson Space Center, 2011. [Online]. Available: <https://curator.jsc.nasa.gov/lunar/lsc/>